



# *MCI Slides*

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# *How to come up with a capital structure for a firm: Main steps*

- 1. Understand the underlying economics of the business
  - Strategic goals
  - Risks
- 2. Determine capital structure target – static trade-off
  - Qualitatively
  - Quantitatively
- 3. How do we reach target? – dynamic trade-offs
  - Cost of straying from target?
  - Lower-cost ways to get to target?
- 4. Evaluate alternatives in a pro-forma model
  - More pragmatic approach than in Diageo
  - “Optimal” capital structure



*Poll: what alternative would you choose?*

- A. \$500M of debt @14.5%, 20yr maturity.
  - B. \$400 million of common stock.
  - C. \$600 million of 9.625%, 20 yr convertibles.
    - Conversion price \$54/share. Callable in 5 years.
  - D. Wait and do nothing.
- 

# *MCI past performance*

|                                 | FY 1974 | FY 1975 | FY 1976 | FY 1977 | FY 1978 | FY 1979 | FY 1980 | FY 1981 | FY 1982 | FY 1983 | Latest  |
|---------------------------------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| Sales growth                    |         | 871%    | 318%    | 121%    | 18%     | 29%     | 52%     | 62%     | 116%    | 112%    |         |
| Operating income / Sales        | -2143%  | -254%   | -37%    | 28%     | 34%     | 32%     | 26%     | 22%     | 33%     | 27%     |         |
| Net income / Sales              | -2871%  | -569%   | -96%    | -3%     | 7%      | 7%      | 9%      | 9%      | 17%     | 16%     |         |
|                                 |         |         |         |         |         |         |         |         |         |         |         |
| Net debt to book capital        | 72%     | 117%    | 124%    | 124%    | 116%    | 94%     | 71%     | 65%     | 55%     | 34%     |         |
| = (D-C)/(D+BE-C)                |         |         |         |         |         |         |         |         |         |         |         |
| Interest coverage = OI/Interest | neg     | neg     | neg     | 1.0     | 1.2     | 1.3     | 1.5     | 1.9     | 4.8     | 5.5     |         |
|                                 |         |         |         |         |         |         |         |         |         |         |         |
| Avg stock price                 | \$3.0   | \$1.5   | \$1.5   | \$1.5   | \$1.8   | \$2.5   | \$3.3   | \$4.8   | \$12.5  | \$31.5  | \$47.0  |
| P/E ratio                       | neg     | neg     | neg     | neg     | 35      | 63      | neg     | 53      | 14      | 19      | 28      |
|                                 |         |         |         |         |         |         |         |         |         |         |         |
| MV equity                       | \$81    | \$46    | \$62    | \$60    | \$71    | \$109   | \$214   | \$359   | \$1,218 | \$3,692 | \$5,508 |
| Net debt to market capital      | 37%     | 68%     | 71%     | 73%     | 70%     | 61%     | 48%     | 43%     | 20%     | 10%     | 7%      |

| <i>Date</i> | <i>Instrument</i>                                                                                                                     | <i>MCI Price<br/>on Issue Date</i> | <i>Amount/Price</i>                         | <i>Net Proceeds</i> | <i>Date Called for<br/>Conversion</i> |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|---------------------------------------------|---------------------|---------------------------------------|
| June 1972   | Common stock                                                                                                                          | IPO                                | 6,000,000 shares<br>@ \$5                   | \$ 27,070,000       | na                                    |
| Nov. 1975   | Common stock plus 5-year<br>warrant attached (exercise<br>price—\$1.25)                                                               | $\frac{7}{8}$                      | 9,600,000 units<br>@ \$1                    | \$ 8,165,000        | na                                    |
| Dec. 1978   | \$2.64 Convertible cumulative<br>preferred stock (conversion<br>price—\$2.1875 per share of<br>common)                                | $1\frac{7}{8}$                     | 1,120,000 shares<br>@ \$25                  | \$25,760,000        | March 1980                            |
| Sept. 1979  | \$1.80 Senior convertible<br>cumulative preferred stock<br>(conversion price—\$5 per<br>share of common)                              | $3\frac{1}{4}$                     | 4,500,000 shares<br>@\$15                   | \$63,125,000        | May 1981                              |
| July 1980   | 15% Subordinated debentures<br>due August 1, 2000                                                                                     | —                                  | \$52,500,000<br>@ 100% of face<br>value     | \$50,545,000        | na                                    |
| Oct. 1980   | \$1.84 Cumulative convertible<br>preferred stock (conversion<br>price—\$9 per share of<br>common)                                     | $6\frac{1}{16}$                    | 3,300,000 shares<br>@ \$15                  | \$ 46,725,000       | Nov. 1981                             |
| April 1981  | 14 $\frac{1}{8}$ % Subordinated<br>debenture due April 1, 2001                                                                        | —                                  | \$125,000,000<br>@ 84.71% of<br>face value  | \$102,055,000       | na                                    |
| Aug. 1981   | 10 $\frac{1}{4}$ % Convertible<br>subordinated debenture due<br>August 15, 2001 (conversion<br>price—\$12.825 per share of<br>common) | $10\frac{7}{8}$                    | \$100,000,000<br>@ 100% of face<br>value    | \$98,200,000        | Feb. 1983                             |
| May 1982    | 10% Convertible subordinated<br>debenture due May 15, 2002<br>(conversion price—\$22.50<br>per share of common)                       | $18\frac{5}{8}$                    | \$250,000,000<br>@ 100% of face<br>value    | \$245,925,000       | Dec. 1982                             |
| Sept. 1982  | 12 $\frac{7}{8}$ % Subordinated<br>debenture due Oct. 1, 2002                                                                         | —                                  | \$250,000,000<br>@ 85.625% of<br>face value | \$209,922,500       | na                                    |
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# *MCI: What happened?*

- ❑ Initially, the offer from Drexel was:
  - ❑ \$ 400 MM in equity or \$600 MM in convertibles
  - ❑ MCI was going to go with the equity
- ❑ Then the phone call came from their investment banker, Michael Milken:
  - ❑ We can arrange debt & warrants for \$1 BILLION.
  - ❑ Too tempting... MCI took the offer
- ❑ Poor AT&T is forced to cut prices by FCC
  - ❑ First by 14%, then by another 3.5%.
- ❑ MCI started out being 30% cheaper...
  - ❑ then 8% cheaper...
  - ❑ and finally only 5% cheaper than AT&T.
  - ❑ Results?

|             | 84   | 85   | 86   | 87   | 88   |
|-------------|------|------|------|------|------|
| SALES       | 1665 | 1959 | 2542 | 3591 | 3940 |
| PROJ. SALES | 1850 | 3160 | 4870 | 7380 | 8660 |
| MY BAD CASE | 1480 | 2528 | 3896 | 5904 | 6928 |
| NET INC.    | 181  | 109  | 68   | 16   | 70   |
| PROJ. NI    | 210  | 235  | 371  | 588  | 731  |
| MY BAD CASE | 81   | 30   | 77   | 150  | 227  |
| CAPX        | 890  | 1157 | 999  | 1072 | 577  |
| PROJ CAPX   | 890  | 1467 | 1881 | 2710 | 1357 |
| STOCK PRICE | 28   | 15   | 22   | 12   | 18   |

- ❑ AT&T got 70% - 80% of subscribers, and most big accounts.
  - ❑ MCI got about a 10% share. GTE Sprint a 7% share.
- ❑ MCI ran out of money in Spring of 1985.
  - ❑ They tried to find a buyer, but could not.
- ❑ Finally, in June 1986, IBM bought 47 million new shares (18% stake) in MCI at a price of \$14 / sh.
  - ❑ Instead of cash, IBM paid for shares by transferring its unprofitable Satellite Business Systems (SBS) division,
    - ❑ SBS owned capital equipment that was MCI could use.
    - ❑ If MCI does well, IBM makes money on their shares.
    - ❑ If MCI goes bankrupt, SBS would anyway have been a write-off for IBM.
- ❑ CEO McGowan had a heart attack and then a transplant.
  - ❑ One MCI exec quoted as saying: "I thought we were a goner."
- ❑ By August 1988, MCI had not converted any debt, and warrants expired unexercised.

- ❑ In 1988, MCI turned the corner.
  - ❑ They exchanged preferred stock for IBM's equity stake.
    - ❑ They paid IBM \$ 29 / share. IBM made roughly  $\$15 \times 47 \text{ M} = \text{or } \$700 \text{ M}$ .
  - ❑ 1988, MCI had positive cash flow for first time ever.
  - ❑ By early 1990's, MCI was a main player in LD phone business.
- ❑ BUT: They would have been a stronger competitor if they had issued equity in 1983 instead of the debt and warrants.
  - ❑ MCI could not raise money when profit margins were slashed.
  - ❑ Had to give \$700 M to IBM.
- ❑ From a low of \$10, stock price reached \$100/share in 1995 .
  - ❑ Over the period, 1983-1995 MCI stock was up 2x (\$ 47 to \$ 100)
  - ❑ S&P500 index was up 5x over the same period.
- ❑ MCI was acquired by Worldcom in 1998 for \$40 billion.
  - ❑ Worldcom went bankrupt in 2002 following an accounting scandal

## ***MCI: WHY THIS CASE?***

- ❑ Comprehensive capital structure analysis
  - ❑ Static trade-off
  - ❑ Dynamic concerns and implementation
- ❑ Convertibles as a financial management tool

# ***Steps in making a capital structure decision***

- (1) Understand the the business and economic environment
  - Qualitatively and quantitatively (historical ratios, S&U)
- (2) Qualitatively evaluate net benefits of debt
  - $PVTS + PV(Inc.) - PV(COFD)$
- (3) Pin these insights down to a target leverage ratio
  - Look at ratios for comparable companies
  - Look at debt ratings
  - Estimate future cash needs in a cash flow statement (S & U)
- (4) Consider dynamic tradeoffs
  - How do we get to target? Any reasons for temporarily deviating?
- (5) Analyze cap. structure policies using projected financials and scenario analysis
  - Focus on being able to handle bad case scenario. “Stress test”

# *Static trade-off theory*

## ■ Costs of debt

- Present value of the costs of financial distress
- Function of probability of distress ( $\text{Pr}(\text{FD})$ ) as well as costs if distress occurs (COFD)

## ■ Benefits of debt:

- Present value of the debt tax shield:  $\text{PV}(\text{TS})$
- Value of the incentive benefits of debt:  $\text{PV}(\text{Inc})$

## ■ Goal is to pick a capital structure that maximizes the value of the levered firm:

- $V_L = V_U + \text{PV}(\text{TS}) + \text{PV}(\text{Inc.}) - \text{PV}(\text{COFD})$

# ***Static Tradeoff for MCI:***

## ■ P(FD)?

1. Highly volatile industry situation due to deregulation
  2. MCI small, with relatively illiquid assets.
  3. MCI is undiversified
- ⇒ P(FD) is high.

## ■ COFD:

1. Large, inflexible capital needs.
  2. Reputation with customers may matter.
  3. Illiquid assets - might have to sell at fire-sale prices if liquidate.
- ⇒ COFD is high.
- ⇒ E(COFD) is high!

## ■ Benefits of debt?

1. Tax benefits valuable only in good case scenario.

- Can always leverage up later when some uncertainty about deregulation has been resolved.

2. Managerial incentives to waste cash no great concern

- Large investment needs, no free cash flow.

■ **Trade-off analysis**  $\Rightarrow$  **Need more equity now**

## ■ **But in 5-10 years (if we succeed):**

- Company more profitable, larger, less volatile, more established
- Investment needs (relatively) smaller, generate excess cash flow
- Pr(FD) down, COFD down, PVTs up, incentive benefits up
- $\Rightarrow$  **Optimal to lever up in the future**

## To get to a quantitative target capital structure, get some guidance from comparables

1. *Competitors' ratios:*
  - MCI has relatively high leverage compared to comps (case exhibit 8)
  - But:
    1. Some competitors (e.g. IBM) are mainly in different businesses → not that relevant
    2. Our main competitor, ATT, is *much* larger and already have a network → can handle higher leverage
  - Conclusion: MCI should not be more leveraged than ATT → aim for  $< 40\% D/(D+E)$

## 2. *Debt ratings:*

- If want to ensure a debt rating, rating criteria indicate what ratios a firm should target
- Why do debt ratings matter:
  - Large fraction of bond investors have a preference investing in higher-rated debt
    - » Particularly “Investment Grade” = at least BBB-
    - » Driven by regulation → “Speculative Grade/Junk” (BB+ or below) lead to higher capital requirements (or even prohibited) for certain institutions (insurance co’s, pension funds)
  - Implications:
    - » Fewer investors willing to buy junk bonds
    - » Particularly in “cold” debt markets → junk-rated debt very expensive or even inaccessible

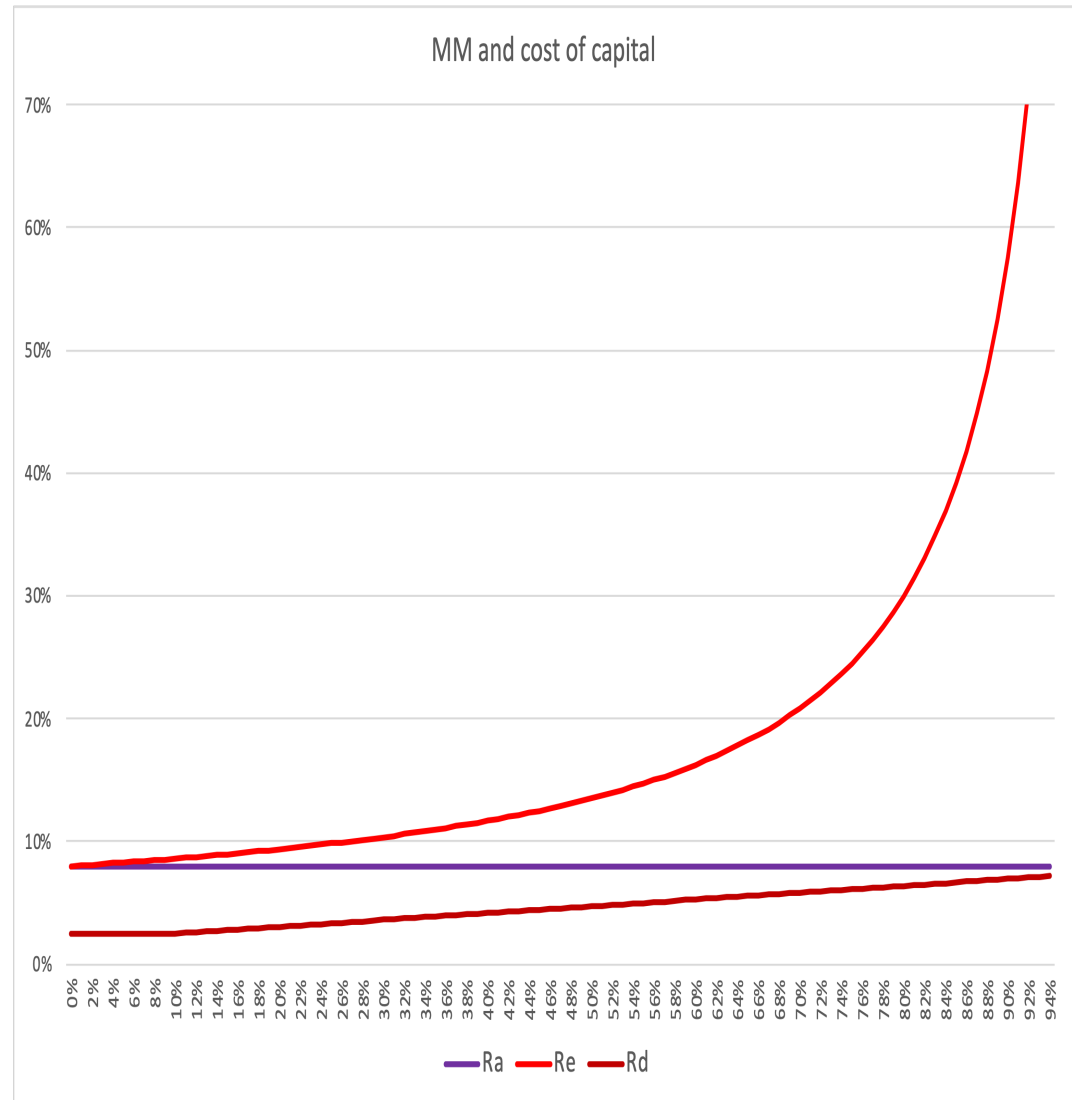
## ■ When/Why, then, do firms need to target a good-enough debt rating?

- If need predictable access to capital markets in the future → should aim for at least investment grade (BBB- or better)

## ■ How important should this be to MCI?

- Very relevant: financing alternatives available today only covers 1-2 years of funding needs → will need to go back and raise more financing in the future

- The fact that interest is higher for lower ratings does *not necessarily* mean a higher overall cost of capital
- Rather: rating is a result of leverage
  - higher risk of default → lower rating
  - higher risk of default →  $R_D$  increases
- $R_D$  always increases with leverage, even when  $R_A$  is constant (Miller-Modigliani Prop II)



# ***Dynamic Issues in Cap Struct. Choice***

- Once you have target capital structure, need to figure out how to get there.
  - If capital markets and managers were "perfect," would always be at target.
  - Still, MCI was unable to reduce leverage throughout 70's. Why?
  
- MCI exemplifies two types of frictions:
  - Signalling costs
  - Debt overhang (see Appendix)
  - Make it difficult to reduce leverage, even though optimal to do so

## ■ A simple example of **signalling costs**

- Firm has 100 shares outstanding, initially trading for \$100/share.
- \$100/share is market's best guess. Management has better info:
  - » probability 1/2, "true value" = \$103.
  - » probability 1/2, "true value" = \$97.
- Management is considering issuing 10 new shares to make an investment.
  - » costs \$970, returns \$990 → worth an extra \$20 to the firm.

## ■ Suppose mgmt knows value is \$103. Will they issue stock?

- Go: sell 10 shares at \$100 not \$103: opportunity cost of \$30.
  - » invest: benefit of \$20.
- No go: do not invest
  - » no cost or benefit.
- Conclusion: With positive info, management
  - » Doesn't issue stock.
  - » Doesn't invest.

- Suppose mgmt. knows value is \$97. Will it issue?
  - Go: sell 10 shares at \$100 not \$97: financing benefit of \$30.
    - » investment benefit of \$20.
  - No go: pass up investment.
  - Conclusion: With negative info, management will issue stock and invest.
- What happens if market observes an equity issue.
  - "Company must be sitting on negative info; otherwise it would not issue."
  - Stock price will decline to \$97.
  - Will make opportunity costs for good firm of issuing even higher: \$60 instead of \$30.
- Dilution problem would be less severe if firms could issue debt for investment
  - Value of senior debt pretty much the same, regardless of whether equity worth +3% or -3%.

- Signaling costs do not necessarily imply that issuing equity will always be costly and should be avoided
  - Equity only costly if firm is undervalued
    - » If firm is actually *overvalued*, these are *good* times to *issue equity*
  - Equity only costly if issuing decision conveys information
    - » Is there a lot of uncertainty about the firm's prospect? Is it likely that management has inside information about this uncertainty? If not, there is no signal from an equity *issue*
  - Is signaling cost  $>$  opportunity cost of not issuing?
    - » If costs of not investing high, or  $E(\text{COFD})$  of issuing debt high, *may still be cheaper to take stock price hit.*

- Lesson: be cautious with asymmetric information arguments against issuing stock: might as well issue if
  - firm actually *is* over-valued
  - uncertainty about firm not *asymmetric* (i.e. equity issue would not signal much to the market)
  
- Did MCI satisfy these conditions?
  - In the late 70' s / early 80' s? Probably.
  - In 1983? Unlikely, given high valuation and recent stock-price run-up.

## *Last step: evaluate proposed capital structure in a pro forma model*

1. Build a pro forma model, and incorporate a “reasonable” bad case scenario
  - Sufficiently likely that it is worth planning for
  - Good enough that it is worth surviving it.
  - (Can always find scenario so bad that best decision is to shut down your business.)
2. Evaluate different capital structures in your pro-forma, and check whether you can avoid financial distress, e.g.
  - Cash to pay debt payments without cutting crucial investments?
  - Meet rating criteria, debt covenants, and competitors’ leverage?
3. Make sure your capital structure allows you to avoid financial distress in the bad case scenario.
  - If PVTs and incentive benefits of debt are important for your firm →, pick the highest debt level that meets this criterion

# ***Convertible Bonds = Hybrid security***

- Convertible debt = Debt + call option on company's equity
- Example:
  - \$1000 bond paying interest of 10%, issued at par
    - » “At par” = convertible is worth exactly \$1000 at issue
    - » Note: 10% < interest on regular debt, since investors also receive a valuable call option
  - Can convert at any time into 20 shares
    - » Conversion Ratio = 20
    - » Conversion Price =  $\$1000/20 = \$50$
  - If stock price  $P > \$50$ , the option is “in the money”
    - » Value of 20 shares > \$1000
  - Still, better for convertible holder not to convert even when  $P > \$50$ 
    - » Convertible can never be worth less than 20 shares
    - » But it is typically worth more, since you get 10% interest as well.
    - » Exception to this rule: company pays large dividend. Needs to own shares to receive this dividend.
  - This is why issuers have the right to force conversion when stock price is high enough
    - » E.g. company has right to buy back convertible for \$1100 when  $P > \$60$
    - » Investor would rather convert since  $20 * \$60 > \$1100$

# ***Convertibles and asymmetric information***

- **Convertibles can mitigate mispricing costs for companies “stuck in the middle”**
  - PV(COFD) is very high; but
  - Equity issue would signal overvaluation → stock price would drop upon announcement.
  - High PV(COFD) → signals to market that company not overvalued → avoids stock price drop and equity dilution
    - » If company were not undervalued it would not expect to be able to convert into equity → would incur COFD → an overvalued firm with high COFD would not issue convertible
- **Convertibles = “Backdoor” equity**
  - Issue convertible, wait until stock price increases (which it should if company undervalued today), and force conversion into equity.

- 
- Note: if it is only equity undervaluation / signaling costs that company worries about, debt would be the least mispriced security.
    - Convertible is more sensitive to firm value because it is a mix of debt and equity → if equity is undervalued, convertible will also be (although less than equity)
  
  - So the key to the argument is that the company has high costs of issuing debt
    - I.e. Pr(FD) and/or COFD
- 

## Appendix:

### Why debt overhang makes equity issues costly

- Consider a firm that has debt outstanding with a face value of \$80 and 10 shares of stock.
- There's a 50% chance the firm will be worth \$100, and a 50% chance the firm will be worth \$20.
- What will the debt and equity trade at (i.e., be worth)?
  - When the firm is worth \$100: the debt receives full value of \$80; the equity receives \$20.
  - When the firm is worth \$20, the debt receives only \$20; the equity gets 0.
  - The debt will trade at \$50 ( $= 1/2 \times 80 + 1/2 \times 20$ ).
  - The equity will trade at \$10 ( $= 1/2 \times 20 + 1/2 \times 0$ ).
    - » Each share of stock will trade at \$1.

- Firm has an investment that requires \$15 up front, but will pay off \$20 for sure. The firm wants to issue equity to finance the investment.
  - Once the investment is financed there is a 50% chance the firm will be worth \$120, and a 50% chance it will be worth \$40.
- Under this situation, what will the debt and equity trade at (i.e., be worth)?
  - When the firm is worth \$120: the debt receives full value of \$80; the equity receives \$40.
  - When the firm is worth \$40, the debt receives only \$40; the equity gets 0
  - The debt will now trade at \$60 ( $= 1/2 \times 80 + 1/2 \times 40$ ). It will increase in value by \$10.
  - The equity will trade at \$20 ( $= 1/2 \times 40 + 1/2 \times 0$ ).

- But remember, the equity includes both old and new shares.
  - There are 10 old shares. The number of new shares equals the number that is necessary to raise \$15 to finance the investment.
  - Because the total equity value is \$20, \$15 represents 75% of the total number of shares.
  - Hence, the new shares must be 75% of the total or no one would buy the equity.
    - » The 10 old shares must control only 25% of the total or \$5. Each old share will trade at only \$0.50.
- By issuing equity to finance a positive NPV investment, this firm will succeed in cutting the value of its old shares in half!
- When will an equity issue make sense, and not "knock the props out from under the stock price?"
  - No debt overhang, i.e., debt trades close to par.
  - Present value of the investment exceeds the transfer to the debt.
    - » If the investment cost only \$10, the old shareholders would break even.